

AN ESRI CANADA
TUTORIAL

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Collect hurricane damage assessments

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1.0 Tutorial Summary

After a hurricane, it's important to assess the damage. In this tutorial, you'll use several ArcGIS apps to build a system to collect and share damage assessments of residential buildings. You'll use Halifax as an example area, but this workflow could be reproduced anywhere. You'll use ArcGIS QuickCapture, ArcGIS Survey123, ArcGIS Field Maps, and ArcGIS Dashboards in this workflow. This tutorial is part of a larger tutorial on collecting hurricane damage assessments in Houston, TX, built by Esri Inc.

Note: If you need to create an operational hurricane damage assessment response, use the [Damage Assessment Solution](#). This tutorial focuses on how users can work together using ArcGIS apps, and the steps described here form a subset to be considered for a damage assessment response.

1.1 Requirements

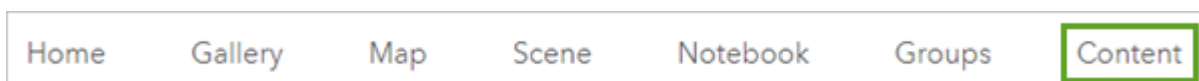
- Creator or Administrator role in an ArcGIS organization
- ArcGIS QuickCapture
- ArcGIS Survey123 Connect
- ArcGIS Survey123 field app
- ArcGIS field maps
- ArcGIS Dashboards

2.0 Configure a feature layer

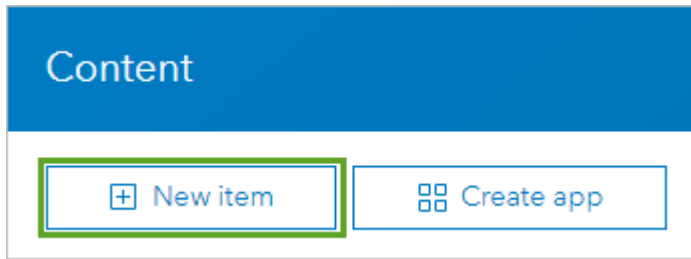
2.1 Create a layer

To collect hurricane damage assessments using various apps, you'll create a project, a survey, and a web map so mobile workers can collect information about damaged buildings. First, you need to create an empty feature layer to store the data that the mobile workers collect. To create the layer, you'll start with a template and add a custom field. You'll also create a view of this layer that will ensure that data cannot be deleted by mobile workers.

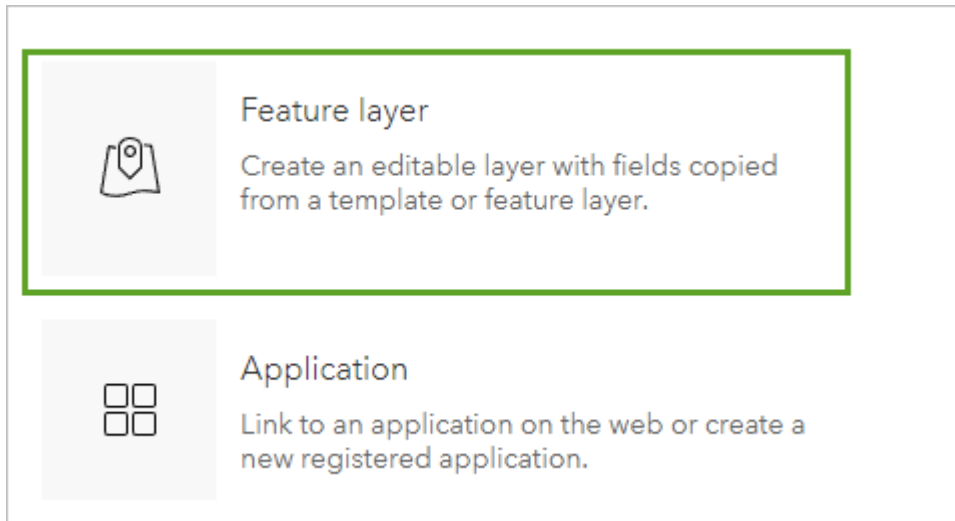
1. Sign in to your ArcGIS organizational account.
2. On the ribbon, click **Content**.



3. On the **Content** page, click **New item**.



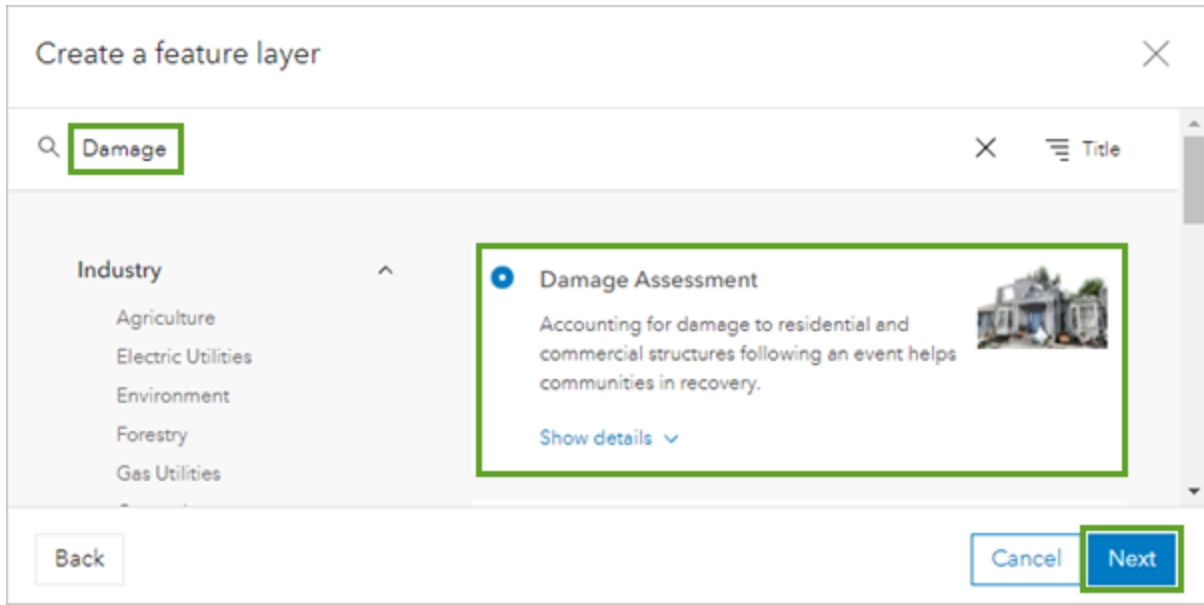
4. In the **New item** window, click **Feature layer**.



5. On the Select an option to create a feature layer page, choose **Use a template** and click **Next**.

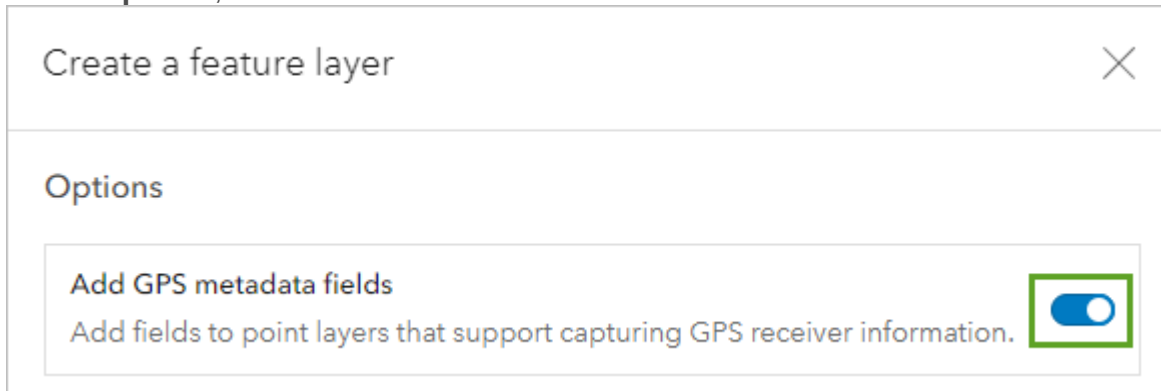
The **Create a feature layer window** appears. It contains many feature layer templates. You'll use the Damage Assessment template and add the custom field later.

6. In the search bar, type *Damage*. Click the **Damage Assessment** feature layer template.



A pane appears with more information about the template.

7. Click **Next**.
8. Under **Options**, turn on **Add GPS metadata fields**.



9. Click **Next**.

10. Enter the following parameters:

For **Title**, type Damage Assessment (and add your name or initials).

For **Tags**, add the tags QuickCapture, Survey123, Field Maps, pressing Enter after each tag.

For **Summary**, type A layer used to collect damage assessments using QuickCapture, Survey123, Field Maps.

Optionally, for **Save in folder**, choose a folder.

New item

Title

Damage Assessment YN

Folder

Your folder

Categories

Assign categories

Tags

Emergency Management

Damage Assessment

State Government

Local Government

Assessment

QuickCapture

Survey123

Field Maps

Add tags

Summary

A layer used to collect damage assessments using QuickCapture, Survey123, Field Maps.

Characters left: 1963

11. Click **Save**.

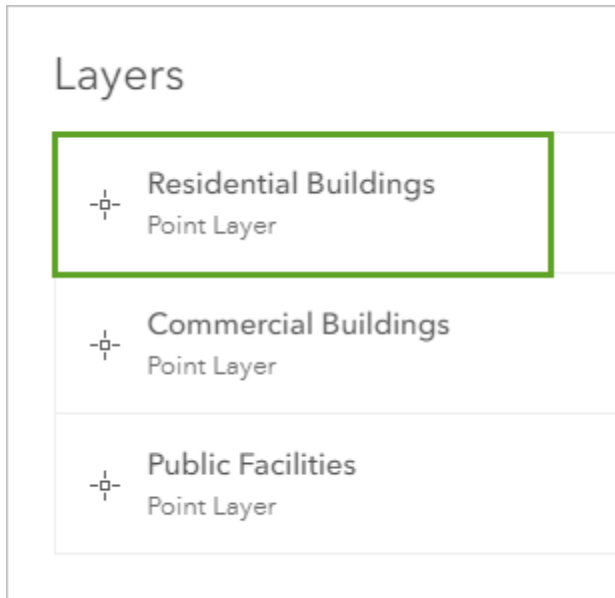
After a few moments, the layer is created and its item details page appears.

2.2 Add a new field

The damage assessment feature layer is the foundation for the work you'll do in this tutorial. It is an empty layer already configured with the fields you'll need while collecting damage assessment data. Your situation requires an additional field that isn't in the default configuration. Because you'll be providing the mobile working teams with various methods for

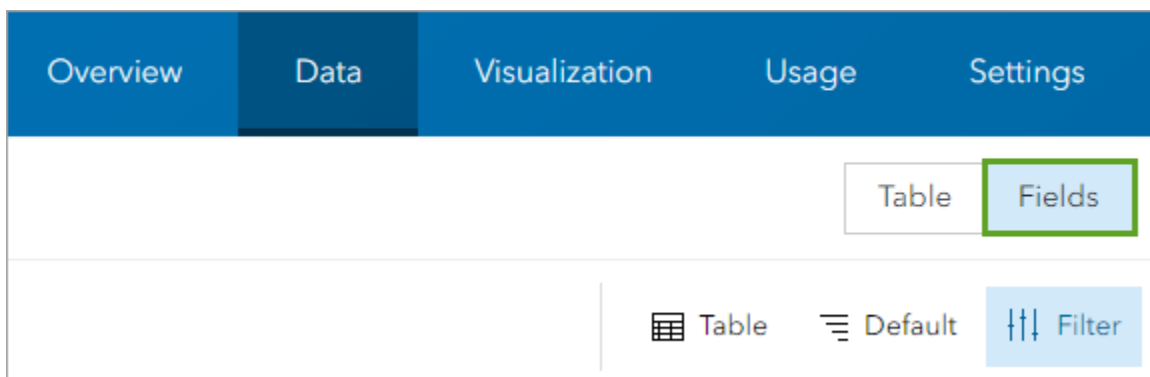
collecting data, you want a field to keep track of how each point was collected. Next, you'll add a field to track this information.

1. If necessary, open the item details page of the **Damage Assessment** feature layer that you just created.
2. On the item details page, under **Layers**, click **Residential Buildings**.



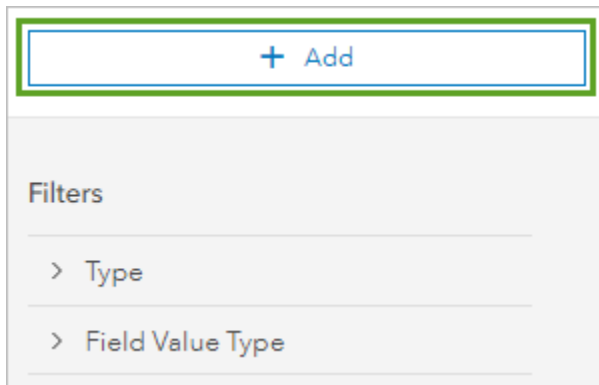
Throughout this tutorial, you'll use the **Residential Buildings** sublayer. The first sublayer is the **Commercial Buildings** layer and is usually shown first whenever you access the layer. Always switch to the **Residential Buildings** sublayer.

3. On the ribbon, click **Data**.
4. Under the ribbon, click **Fields**.



All of the fields provided by the solution are listed. You won't use most of them for this project, but you won't delete them either, in case they are needed by another part of the solution.

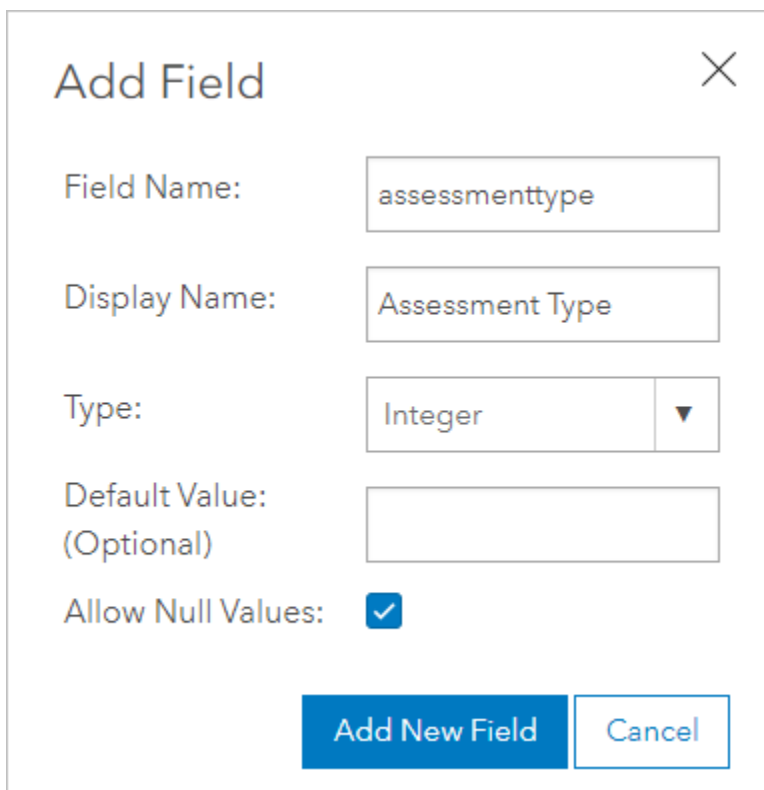
- At the top of the page, click **Add**.



The screenshot shows a user interface with a green-bordered box at the top containing a blue plus icon and the text '+ Add'. Below this is a 'Filters' section with two expandable items: '> Type' and '> Field Value Type'.

The **Add Field** window appears.

- For **Field Name**, type `assessmenttype`.
- For **Display Name**, type `Assessment Type`.
- For **Type**, choose **Integer**.
- Ensure **Allow Null Values** is checked.



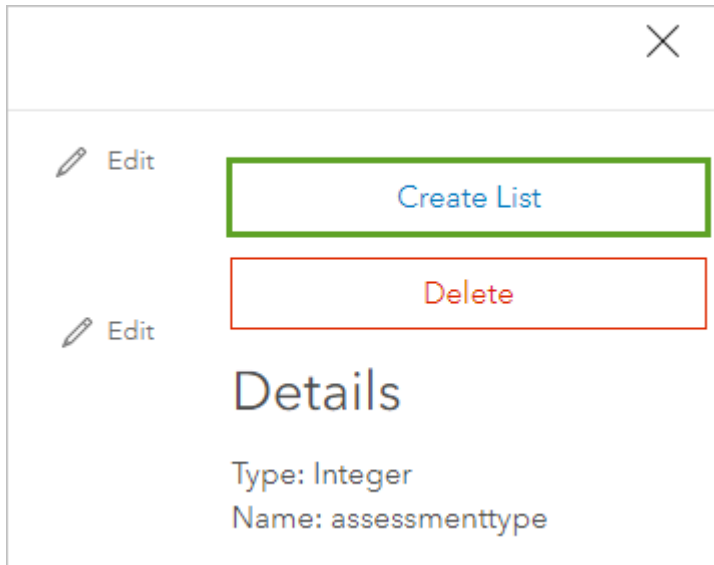
The 'Add Field' dialog box is shown with a close button (X) in the top right corner. It contains the following fields and controls:

- Field Name:** Text input field containing 'assessmenttype'.
- Display Name:** Text input field containing 'Assessment Type'.
- Type:** Dropdown menu with 'Integer' selected and a downward arrow.
- Default Value: (Optional):** Empty text input field.
- Allow Null Values:** A checked checkbox.
- Buttons:** 'Add New Field' (blue) and 'Cancel' (white with blue border) at the bottom.

- Click **Add New Field**.
- Scroll to the bottom of the list of fields. Near the bottom of the list, click **Assessment Type** to view the properties for the new field.

Later in the tutorial, you'll set up three methods for people to collect data. Next, you'll list these three assessment types in the new field.

12. On the **Assessment Type** field page, click **Create List**.



✕

Edit Create List

Edit Delete

Details

Type: Integer
Name: assessmenttype

This list will limit the allowed values for the field. When you use the **DamageAssessment** layer later to collect data, you'll only have three options for the **Assessment Type** field.

13. Under **Label**, type Windshield damage assessment QuickCapture. Under **Code**, type 0.
14. Click **Add**. In the new row, for **Label**, type Preliminary damage assessment Survey123, and for **Code**, type 1.
15. Click **Add**. For **Label**, type Preliminary damage assessment Field Maps, and for **Code**, type 2.

Label	Code	
Windshield damage assessment QuickCapture	0	(0)
Preliminary damage assessment Survey123	1	(0)
Preliminary damage assessment Field Maps	2	(0)

Now, each time data is collected, the method in which it was collected will be stored in the **Assessment Type** field. It will be stored as an integer—0, 1, or 2—but it will display with the labels defined in this list.

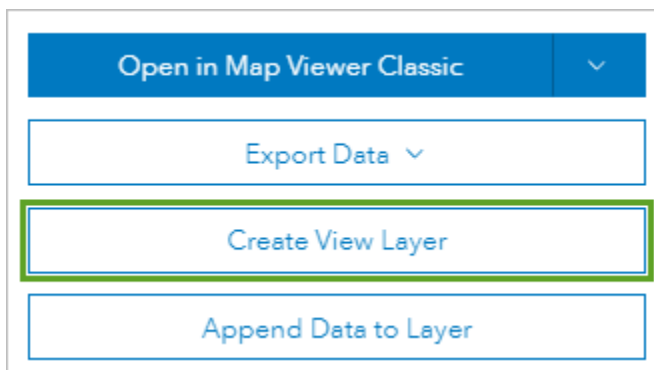
16. Click **Save**.
17. Close any open browser tabs for your ArcGIS organization.

So far in this tutorial, you've created a feature layer and added a new field to track in which app damage assessments are being collected. You also created a list on the new field to limit the allowed values. Next, you'll create a project in ArcGIS QuickCapture, a survey in ArcGIS Survey123, and a web map in ArcGIS Field Maps. These will be used to collect damage assessments at different levels of detail or focus.

2.3 Create a view layer

Because you'll have multiple teams using the same data layer, you'll protect your data by only allowing users to add and update records. They cannot delete records.

1. On the **Contents** page of your ArcGIS organization, select the **Damage Assessment** feature layer.
2. On the item details page for the **Damage Assessment** feature, click **Create View Layer**.



The **Create View Layer** window appears. The first page is **Choose layers** with all four available layers selected.

3. Under **Layers**, uncheck **US National Grid**, **Public Facilities**, and **Commercial Buildings**. The **Residential Buildings** layer is the only one selected.

Damage Assessment YN Source layer	
Layers The layers you select here will be included in the view layer.	
	US National Grid Layer
	Public Facilities Layer
✓	Residential Buildings Layer
	Commercial Buildings Layer

- Click **Next** two times.
- On the map, navigate to Halifax, NS.
The map is now focused on your area of interest.
- On the **Create View** page, for **Title**, type Damage Assessment View and add your initials.

Create Damage Assessment YN	
Title	Damage Assessment View YN
Folder	Your folder

- Click **Create**.
It may take a few minutes for the layer to create. A success message appears and the item details page of the new layer appears.
- On the item details page of the **Damage Assessment View** feature layer, on the ribbon, click **Settings**.

9. On the **Settings** page of this new layer, scroll down to the **Feature Layer (hosted,view)** section. Under **Editing**, check the box for **Enable editing**.
10. Under **What kind of editing is allowed**, uncheck **Delete**.

Editing

☒ Enable editing

☐ Keep track of created and updated features. *

☒ Keep track of who created and last updated features. *

☐ Enable Sync (required for offline use and collaboration).

• Who can edit features?
Share the layer to specific groups of people, the organization

• What kind of editing is allowed?

☒ Add

☐ Delete

☒ Update

☐ Attributes only

☒ Attributes and geometry

[Manage geometry updates](#)

11. Next to **Feature Layer (hosted,view)**, click **Save**.

This feature layer view will be used by each of the field apps to ensure that mobile workers have access to add and update records but not delete them.

3.0 Create a QuickCapture Project and web map for use in Field Maps

3.1 Create a project for windshield damage assessments

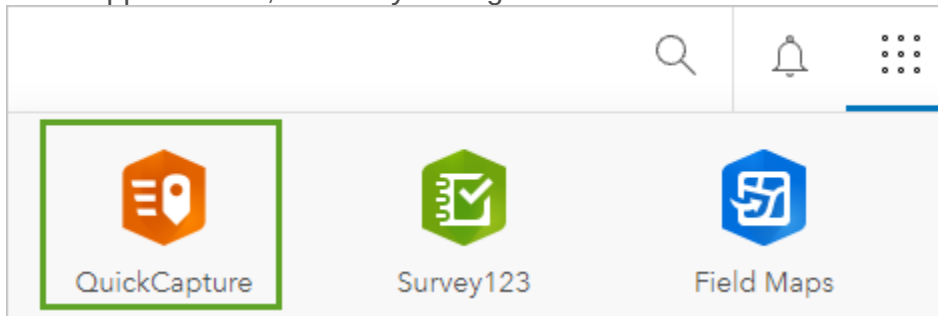
This windshield damage assessment will be used by workers using QuickCapture who first encounter damage so they can flag the building for further assessment. The second part is a preliminary damage assessment, which is used by workers using Survey123 trained to estimate the extent of damage and collect more detailed information. **This second part has already been created for you to save time.** These workers will target damage assessments identified as destroyed by the windshield damage assessment.

Additionally, you'll configure a web map for use in Field Maps. Some teams will focus on damage assessments in a specific area so seeing all records in that location is a priority. Providing the option to collect records with

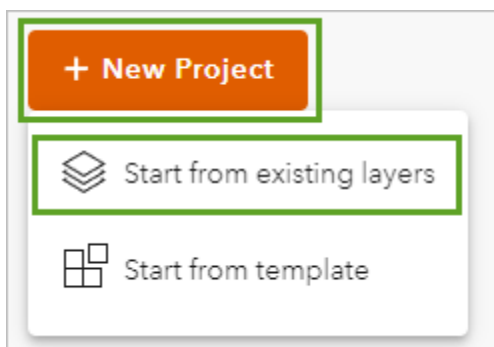
QuickCapture, Survey123, or QuickCapture will give responders the opportunity to use a tool that suits their needs while ensuring everyone is working with the same data.

You'll create a QuickCapture project using the feature layer view. In the project, the different damage types will appear as individual buttons.

1. Launch QuickCapture from the app launcher in your ArcGIS organization. If this app is not in the app launcher, contact your organization administrator.

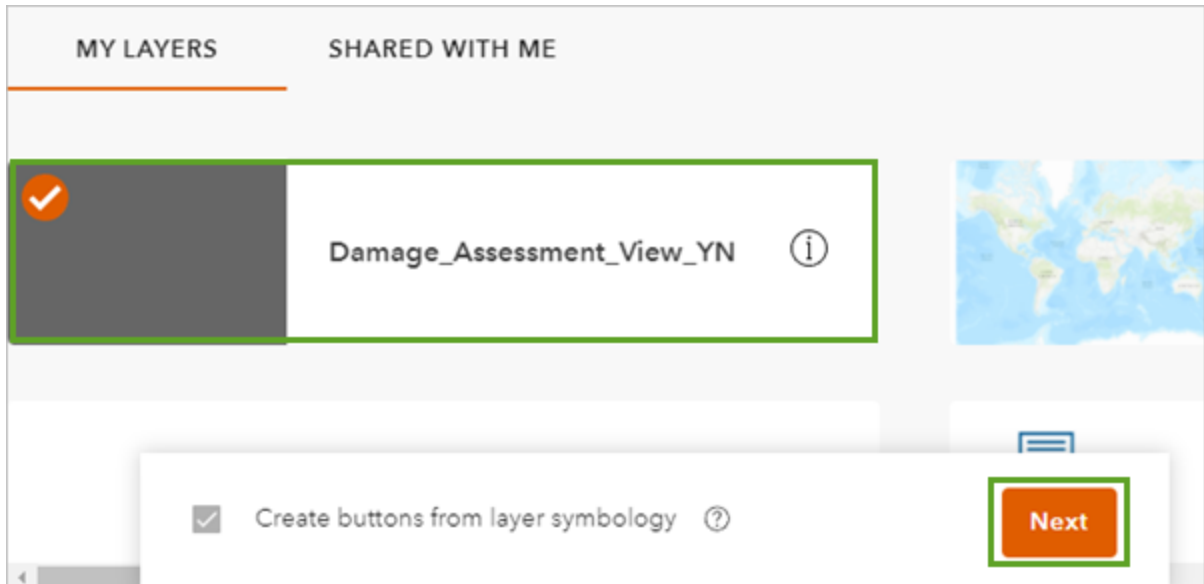


2. Click the **New Project** button and choose **Start from existing layers**.



You're prompted to choose the feature layers you want to use in the project.

3. Locate and click your **Damage Assessment View** feature layer to select it. At the bottom of the screen, click **Next**.

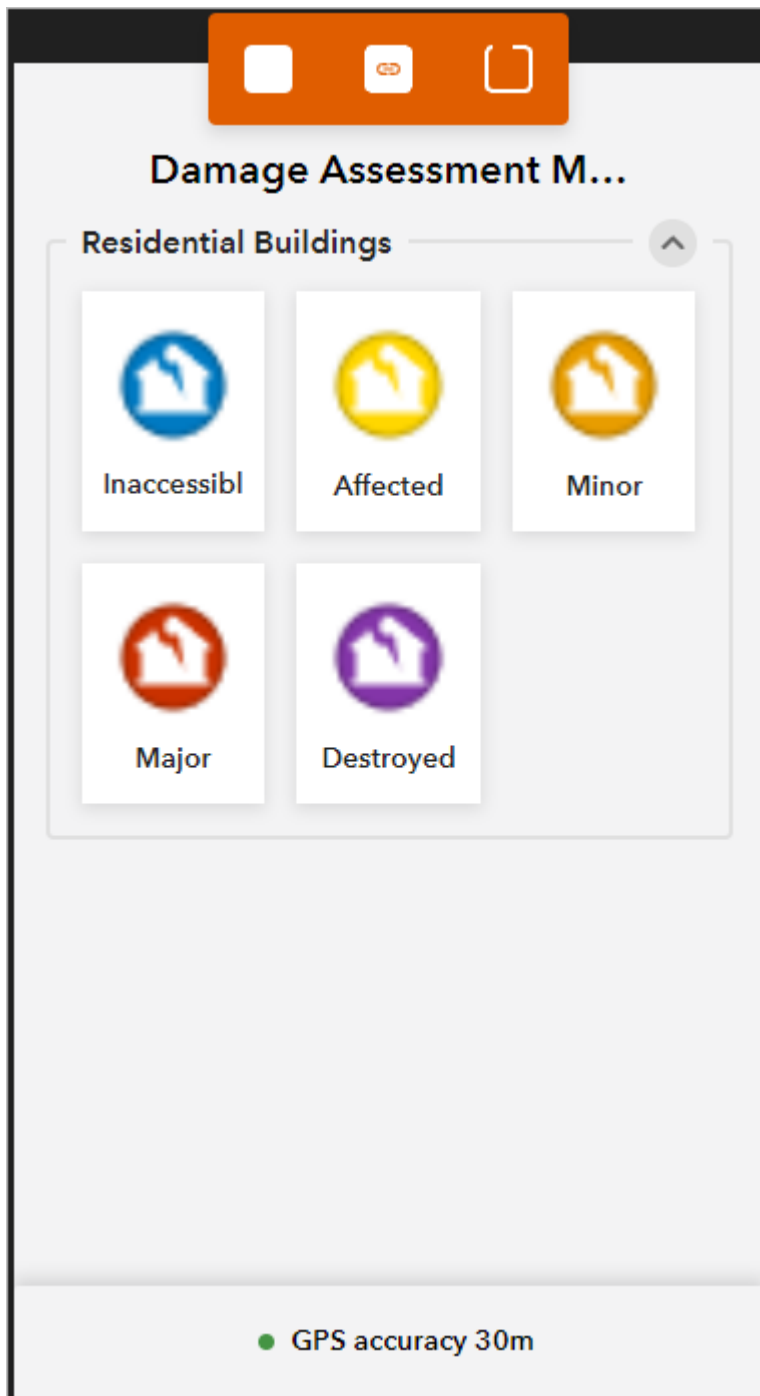


Next, you'll confirm the project's basic information, such as its title.

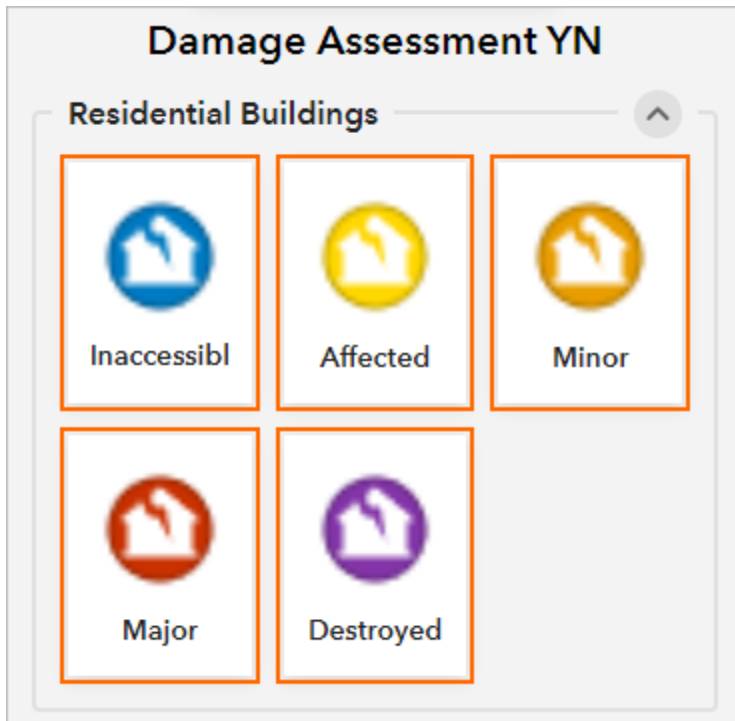
4. For **Title**, type Damage Assessment and add your name or initials. Confirm that **Data recovery email** is set to a valid email address.

A screenshot of the Esri QuickCapture project creation form. The form has three sections: 'Title' with a text input field containing 'Damage Assessment YN'; 'Data recovery email' with a text input field containing 'your_email@domain.com'; and 'Save in folder' with a dropdown menu showing 'Root'. At the bottom right of the form is an orange 'Create' button.

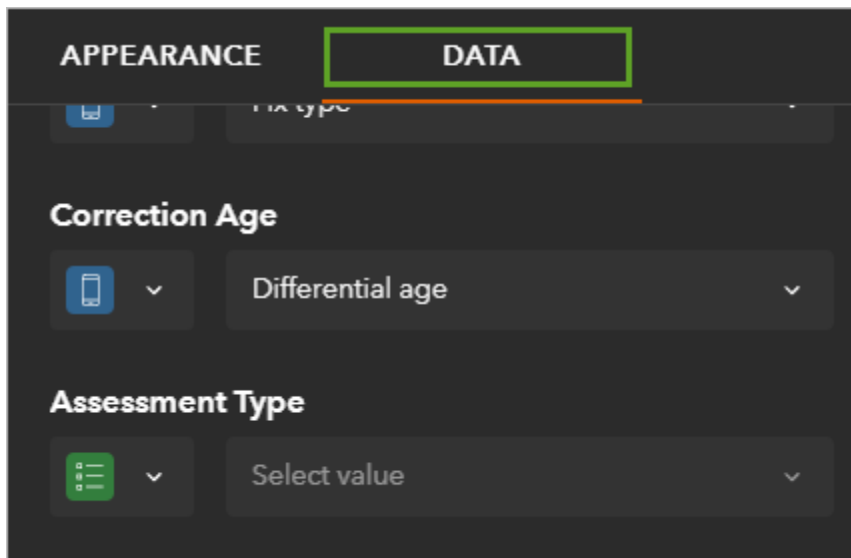
5. Click **Create**.
The QuickCapture project is created. It displays a preview of the app that your mobile workers will access using their mobile devices.



6. In the project preview, select one button. Press and hold the Shift key and click all of the other buttons so that all buttons are selected.



7. In the side panel, click the **Data** tab. Scroll down the list to find the **Assessment Type** field.



8. Click the **Assessment Type** field and choose **Windshield damage assessment QuickCapture** from the list.

Now, every record created in QuickCapture will have the expected assessment type value of 0.

9. On the ribbon, click **Save**.

A message appears, confirming that your project is saved and can be downloaded in the QuickCapture mobile app.

10. Close the QuickCapture project.

11. Optionally, using the QuickCapture mobile app on your personal mobile device, sign into your ArcGIS Account and test out the project you just created.

4.0 Configure a web map for preliminary damage assessment

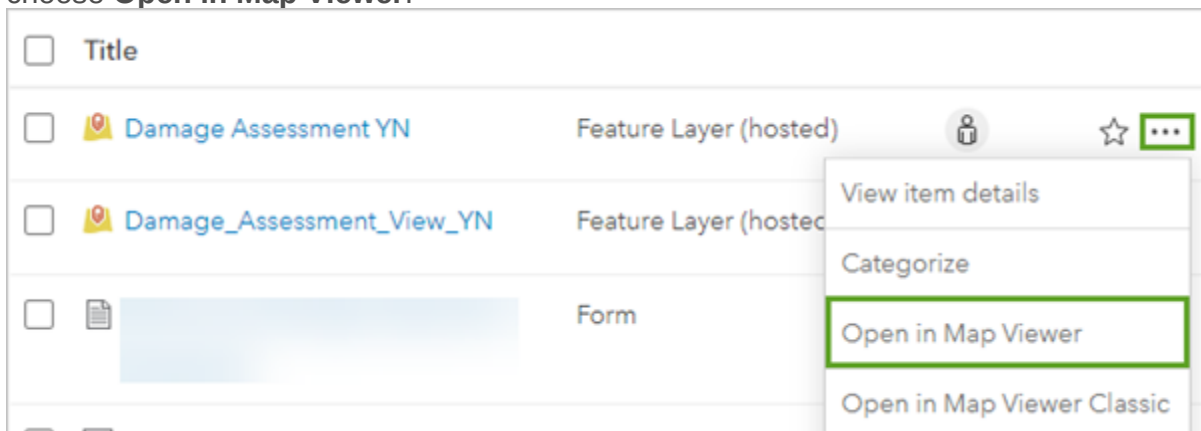
Some teams need to focus on damaged buildings within a specific area of the city, regardless of damage extent. For these teams, a map-based collection method is more useful than a survey. Next, you'll create a web map for use in Field Maps. Field Maps, like Survey123, collects information using mobile devices. But instead of filling out surveys, users add new features and attribute information on a map.

First, you'll make a web map. You'll configure the map's pop-up and you'll share the map so members of your organization can access it.

1. If necessary, sign in to your [ArcGIS account](#).
2. On the ribbon, click **Content**.

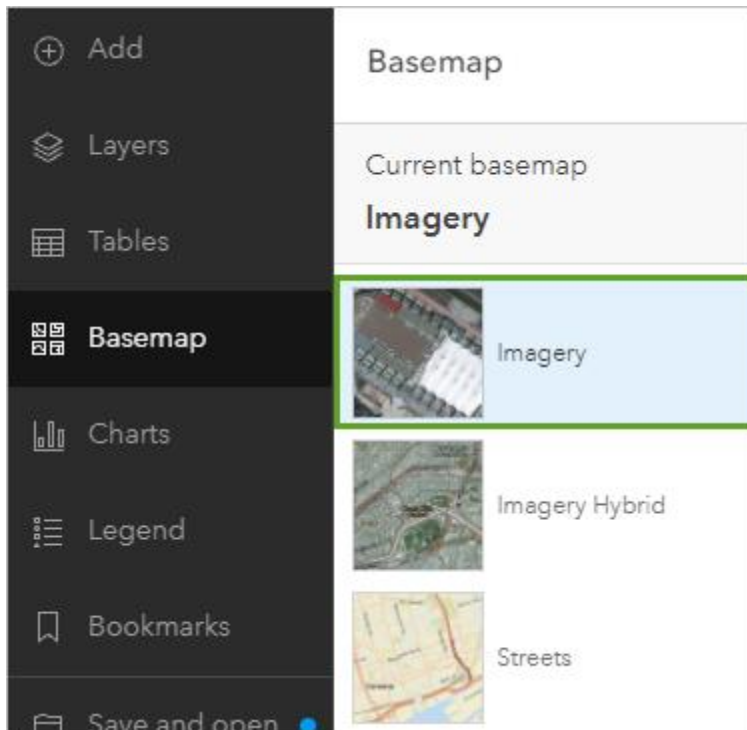


3. Next to the **Damage Assessment View** feature layer, click the options button and choose **Open in Map Viewer**.



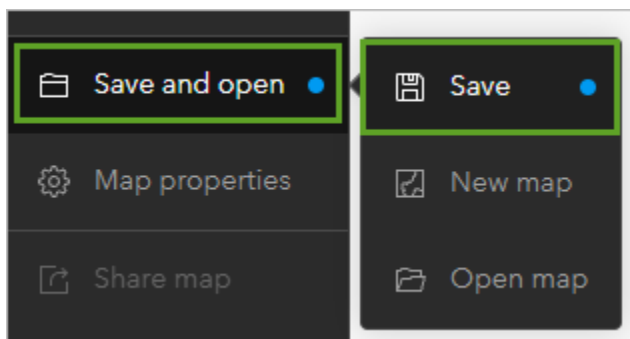
Map Viewer appears. The **Contents** pane shows the sublayers of the feature layer and collected points on the map.

4. On the **Contents** pane, click **Basemap**. On the **Basemap** tab, choose **Imagery**.



The map updates with the basemap.

5. On the **Contents** pane, click **Save and open** tab, click **Save**.



The **Save map** window appears.

6. In the **Save map** window, for **Title**, type Preliminary Damage Assessment Map, add your initials, and click **Save Map**.

Save map

×

Title

Preliminary Damage Assessment Map YN

Tags

Add tag(s)

Summary

Description of the map

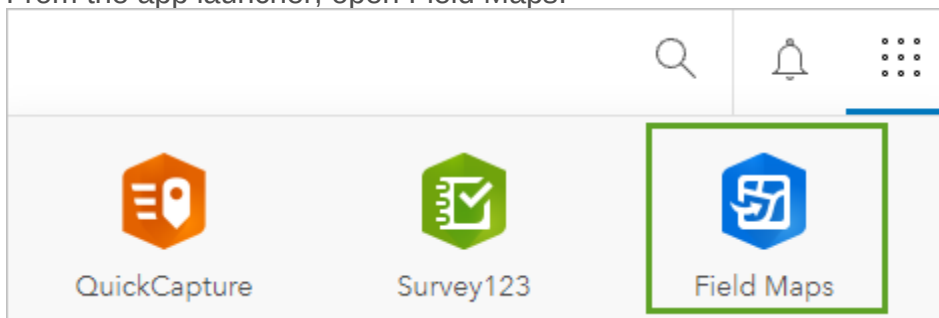
Save in folder

Your folder

Cancel

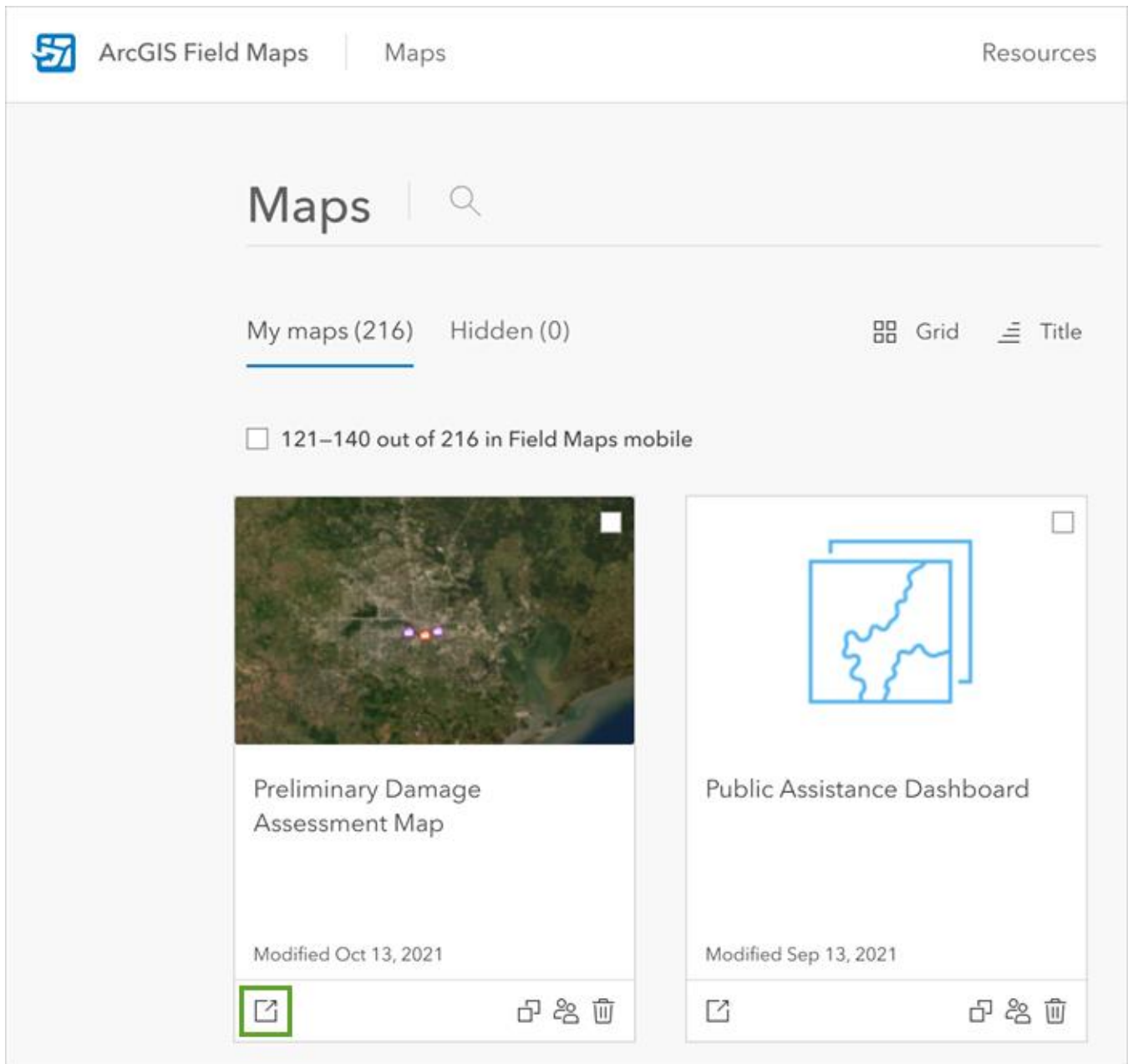
Save map

- From the app launcher, open Field Maps.

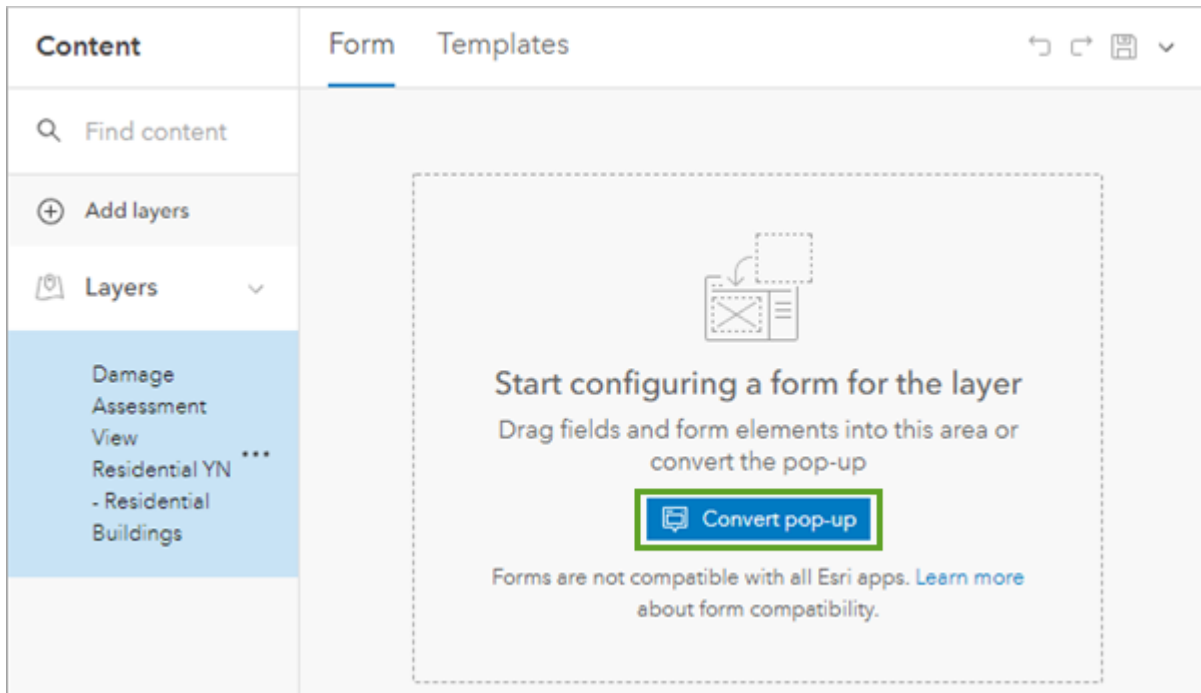


ArcGIS Field Maps web app opens on the **Maps** page.

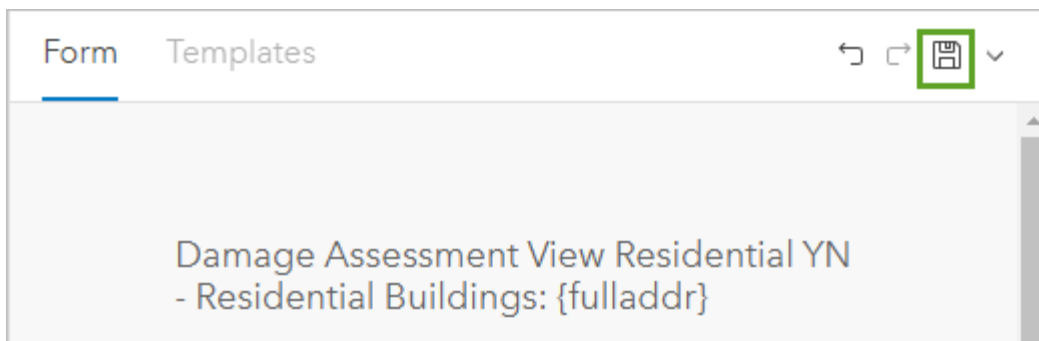
- On the **Maps** page, locate the card for the **Preliminary Damage Assessment Map** you just saved and click **Configure**.



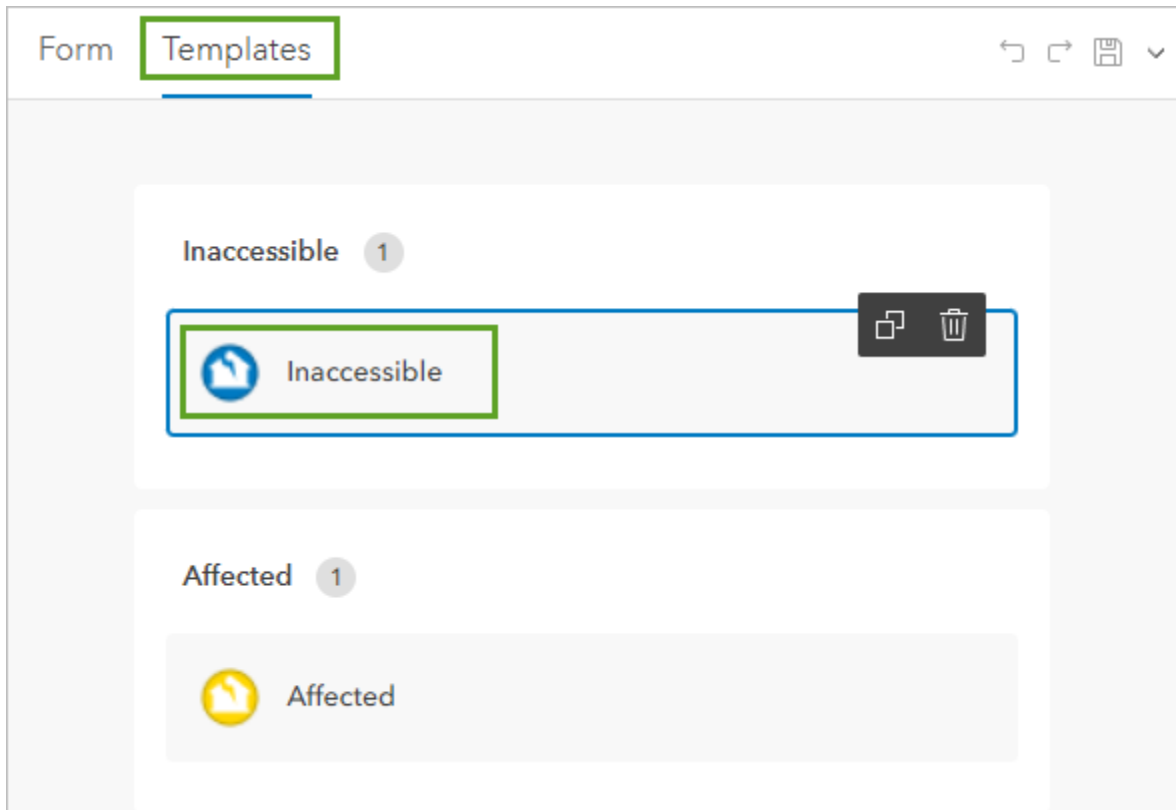
9. Click the **Content** tab.
10. On the **Content** pane, click the **Damage Assessment View - Residential Buildings** layer.
11. In the **Form** pane, click **Convert pop-up**.



12. Click **Save** to save the form.



13. Click the **Templates** tab, select the first feature template.



The **Properties** pane for the feature template appears.

14. In the **Properties** pane, for the **Assessment Type** field, choose **Preliminary Damage Assessment in Field Maps**.

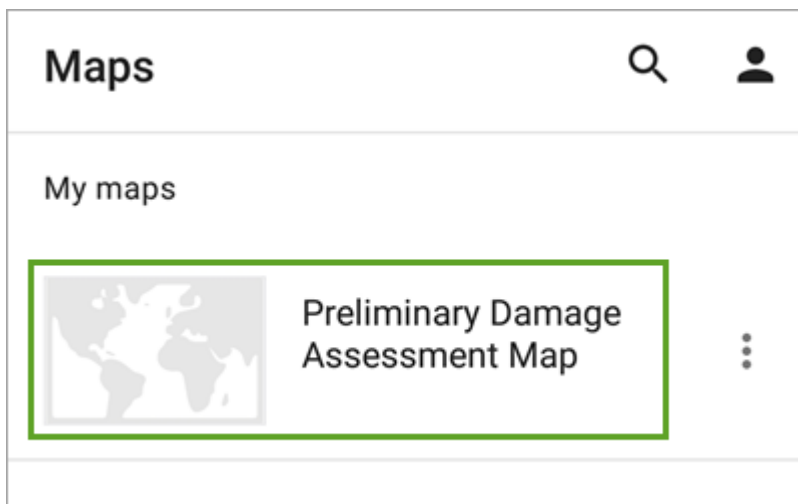
Repeat for each template. When any new records are captured with a feature template in Field Maps, the assessment type will be automatically assigned.

15. Use what you have learned to set **Assessment Type** to **Preliminary Damage Assessment in Field Maps**
16. Click **Save** to save the **Templates**.
17. Close ArcGIS Online.

5.0 Optional: Collect Data

Next, you'll open your web map in Field Maps and collect a sample point to test the collection method

1. Open Field Maps on either your Android or iOS mobile device.
2. Sign in with your ArcGIS account.
3. Click the **Preliminary Damage Assessment Map** to open it.



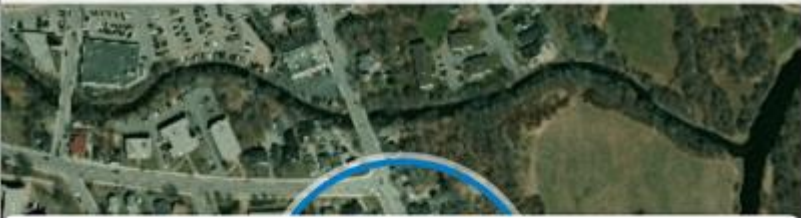
The map is centered on your current location.

4. Click the **Add** button.
5. Pan the map to center it somewhere in the HRM.

A list of possible features appears.

6. Select **Inaccessible** from the **Damage Assessment - Residential Buildings** layer.

GPS accuracy 4.7 m



DamageAssessment - Residential Buildings

-  Affected
-  Destroyed
-  Inaccessible
-  Major
-  Minor

A form appears for the new feature. You can enter some sample information here. You'll start by changing the point's location.

7. Click **Update Point**.
8. Swipe the menu upward to view the form parameters.

The **Assessment Type** parameter is already set to **Preliminary damage assessment Field Maps**.

9. For **Estimated \$ Loss Structure**, type 6000. For **Incident Name**, type Test 2.
10. Click the check mark or **Submit** to collect the new feature.

×

Collect

GPS accuracy 3.3 m

New Feature

29.721753°N 95.331994°W

TAKE PHOTO

ATTACH

Incident Number

Incident Name

Test 2

11. Close Field Maps.

6.0 Preview Survey Created with Survey123 Connect

A preliminary damage assessment survey was created for you to save time. This was created in Survey123 Connect by automatically generating it from the existing feature layer view from above. The survey provides opportunity for users to describe the damage in more detail than

the windshield damage assessment. For this reason, the inbox was also enabled so that users can see records created during the windshield damage assessment and add more information to the existing record.

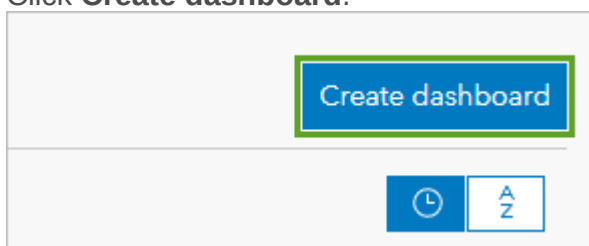
1. Go to the following link:
<https://survey123.arcgis.com/share/38c8723734504b15ae939264902d3727>
2. Scroll through the form to see another method of collecting damage assessments in more detail.
3. To see detailed instructions on creating this form in Survey123 Connect, reach out to Erica at ecorbett@esri.ca

7.0 Create a dashboard

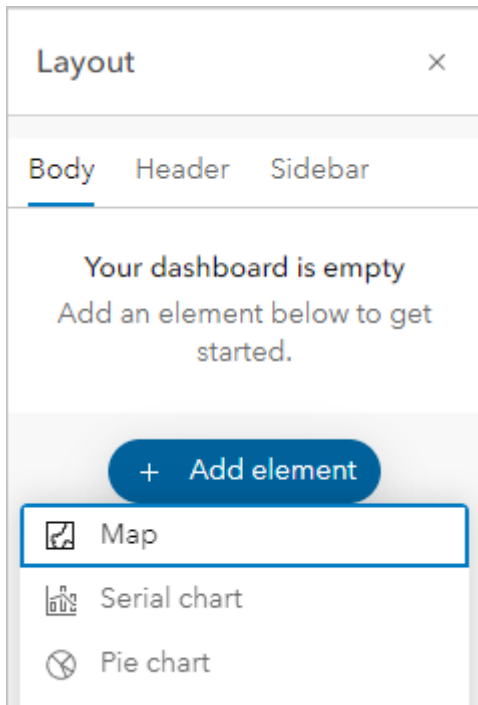
To complete this tutorial, you'll create a dashboard with ArcGIS Dashboards to display all of the collected data. The dashboard will also display a chart that shows the proportion of records captured with each app. Charts and indicators are useful to quickly view progress of the damage assessment workflow. In this tutorial, you will only add one chart, but to see a dashboard with the chart and additional indicators, view the [sample dashboard](#) that also includes hundreds of (simulated) damage assessments (for Houston, TX).

First, you'll create a dashboard from the web map that you created earlier.

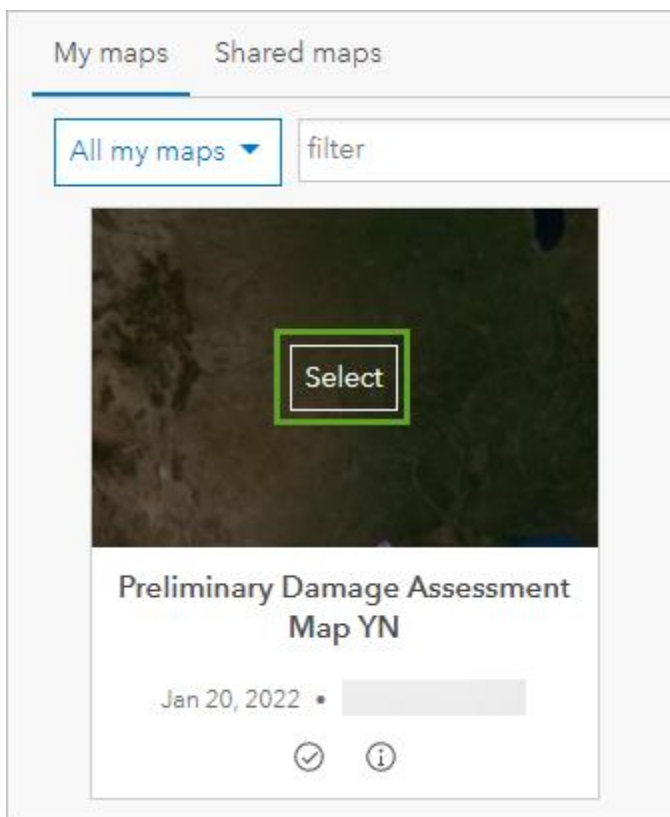
1. Launch Dashboards from the app menu in your ArcGIS organization.
2. Click **Create dashboard**.



3. For **Title**, type Damage Assessments Dashboard. Click **Create dashboard**.
4. On the **Layout** pane, click the **Add element** button and choose **Map**.



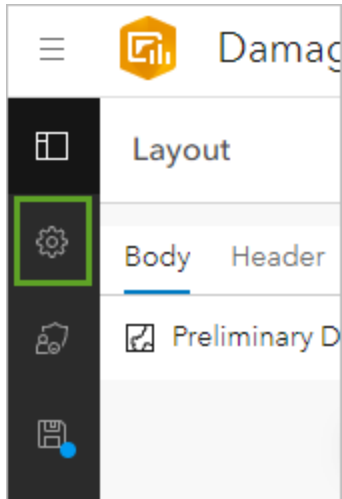
5. From the gallery, point to your **Preliminary Damage Assessment Map** and click **Select**.



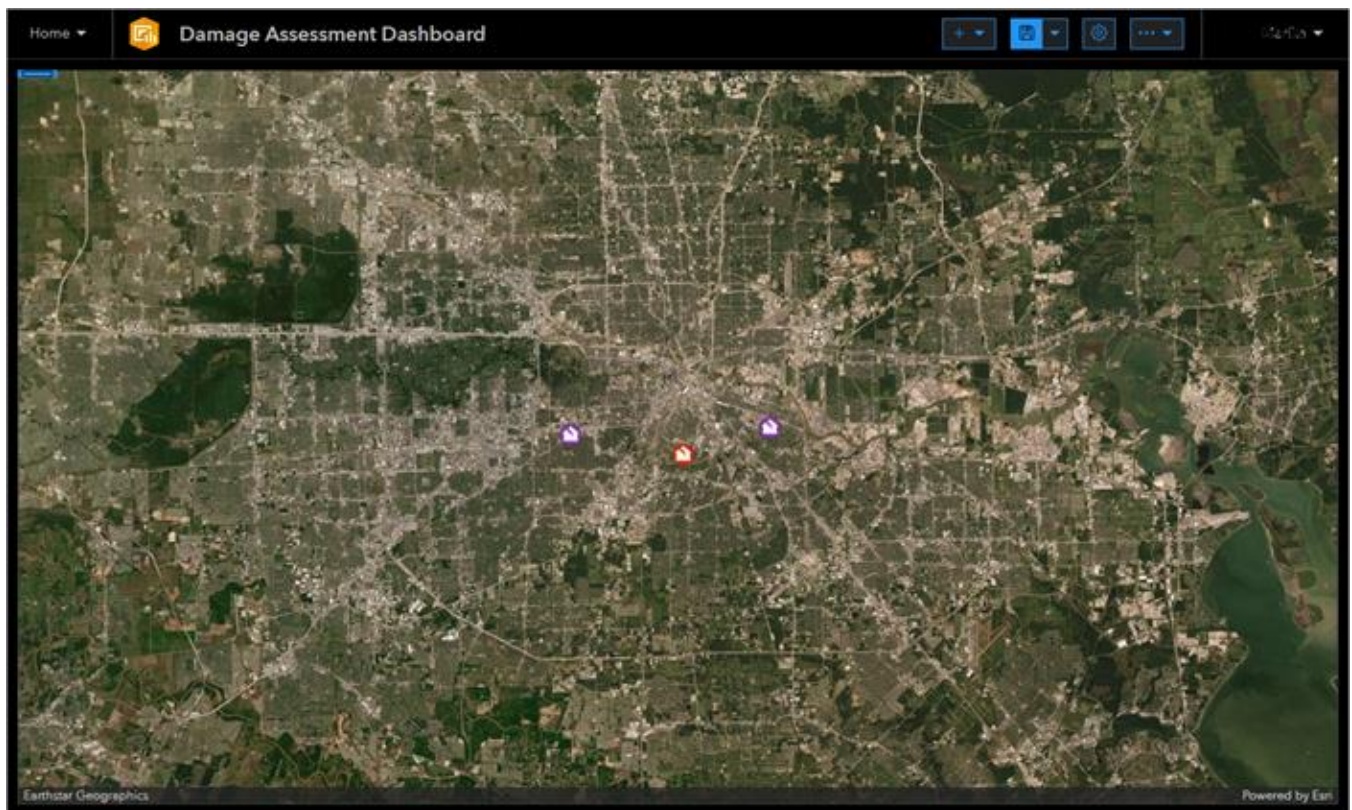
6. Click **Done**.

The map is added to your dashboard, showing the points that you have already collected.

7. On the map, zoom to Halifax (or where you collected your points) so your collected points are more visible.
8. On the dashboard toolbar, click the **Settings** button.



9. Under **Layout**, for **Theme**, choose the **Dark** theme and click **Done**. The dashboard theme updates.



10. On the dashboard toolbar, click the **Save** button and choose **Save**.

The content of the map is readily available for you to choose to display in lists, charts, and indicators.

11. In the **Layout** pane, click the **Add element** button and choose **List**.

12. Choose the **Damage Assessment View** layer.

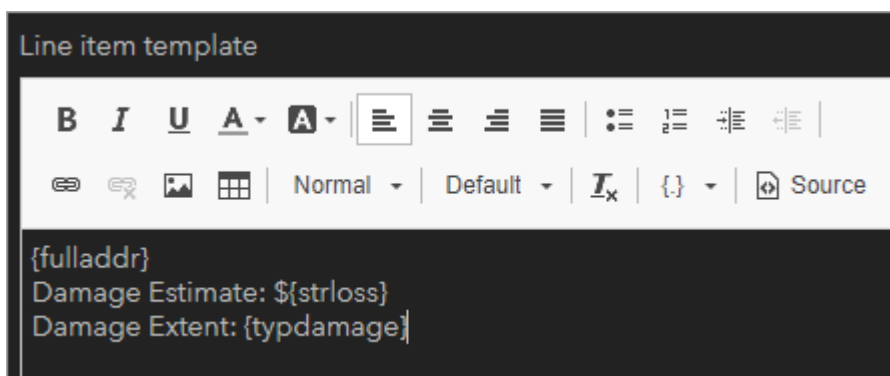
More options become available.

13. Click the **List** tab.

14. In the **Line item template** rich text editor, type the following text:
{fulladdr}

Damage Estimate: \${strloss}

Damage Extent: \${typdamage}



15. Click **Done**.

The list will automatically fill half of the dashboard. Click the vertical line that separates the list from the map and drag it so that the list occupies approximately 30 percent of the dashboard width.

16. In the **Layout** pane, click the **Add element** button and choose **Pie chart**.

First, you'll choose the layer on which the pie chart will be based.

17. Choose the **Damage Assessment View** layer.

More options become available.

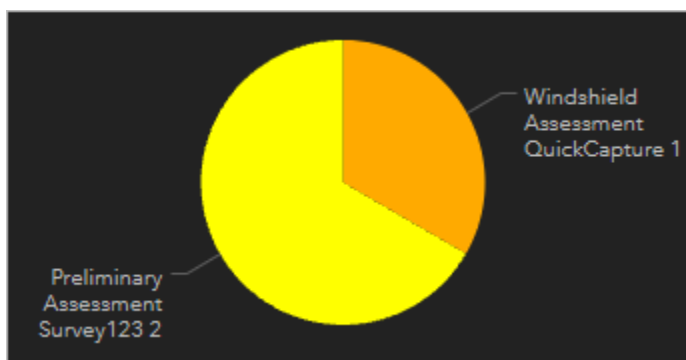
18. On the **Data** tab, for **Category Field**, choose **Assessment Type**.

Filter + Filter

Categories from Grouped values Features Fields

Category field Assessment Type

A pie chart appears in the preview. As your mobile workers add more assessments, the pie chart will update.



The labels make the chart very wide. You'll replace them with a legend.

19. Click the **Chart** tab.
20. In the **Chart options** pane, under **Labels**, for **Visibility**, click **Hide**.
21. Under **Legend**, for **Visibility**, click **Value**. For **Placement**, click **Side**.

Labels

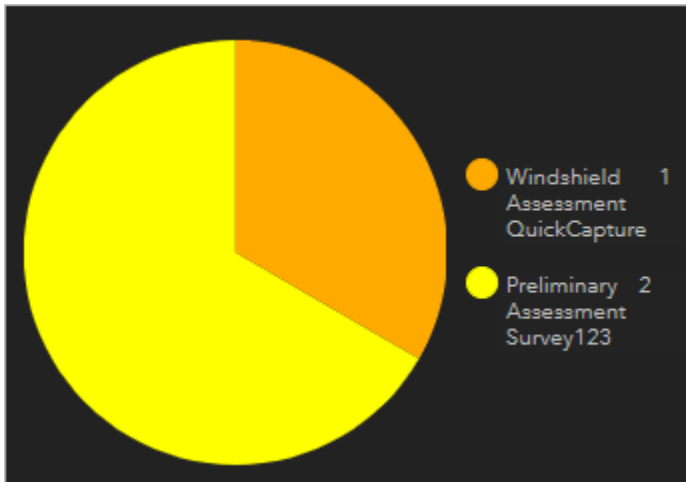
Visibility Hide Value Percentage

Legend

Visibility Hide Value Percentage

Placement Bottom Side

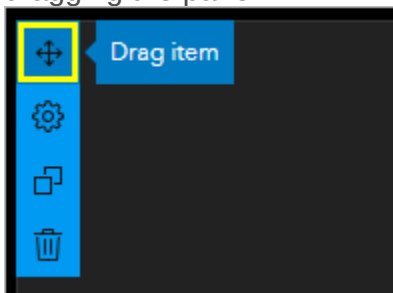
The preview of the pie chart now shows a legend on the side.



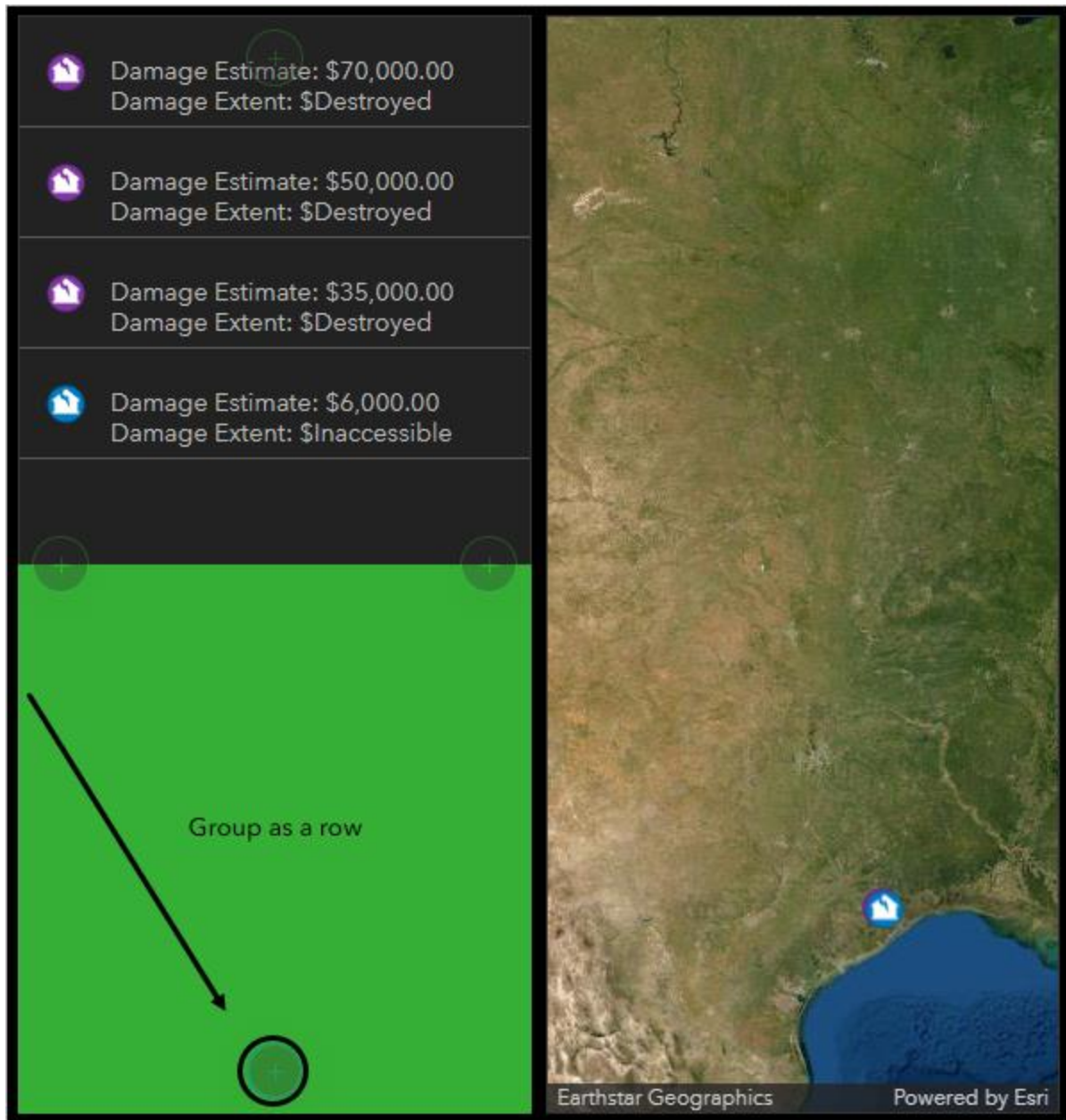
22. Click **Done**.

The pie chart is added to the dashboard, but as a pane that takes up half the dashboard's space. You'll reposition the pie chart to below the list.

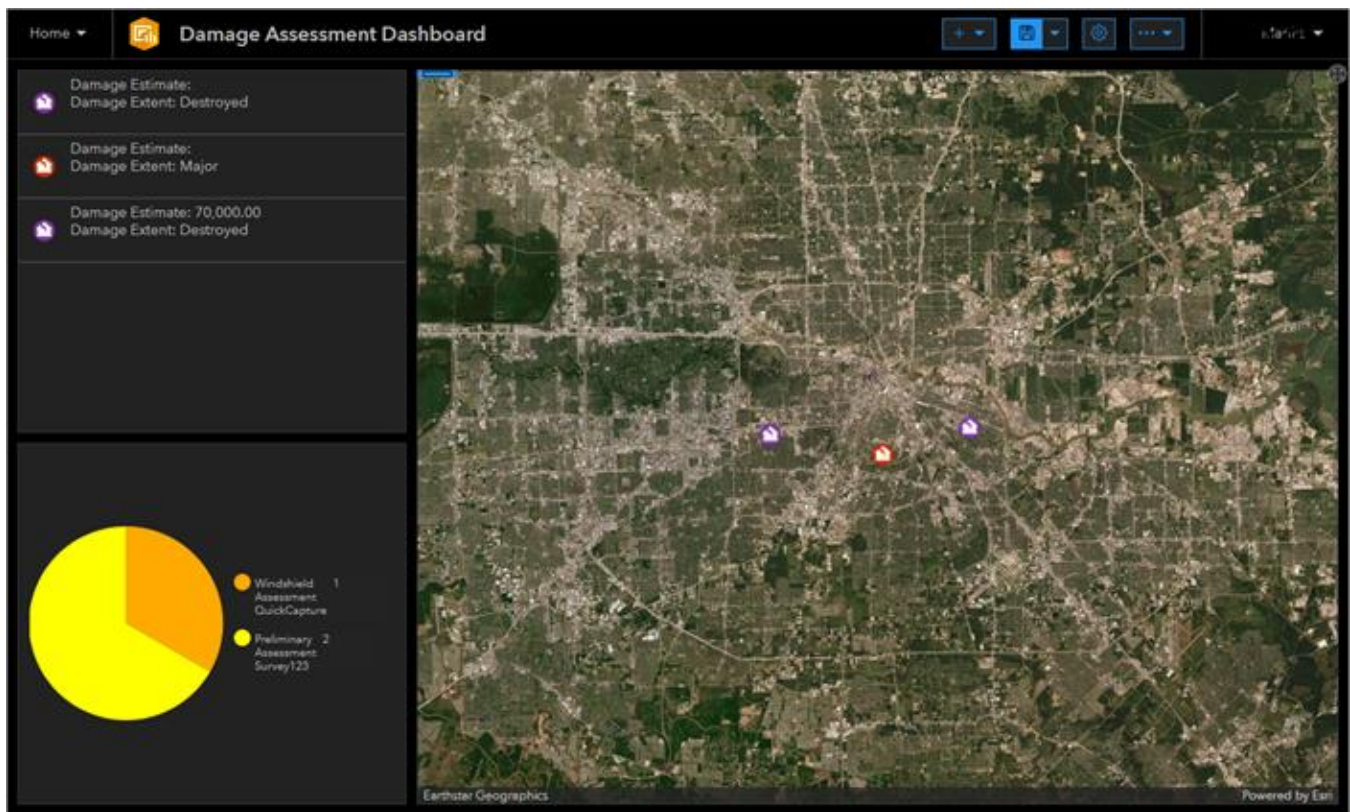
23. Point to the upper corner of the pie chart pane element and click the **Drag item** button while dragging the pane.



24. While dragging the pane, press Shift to show pane groupings. Place the pie chart under the list element.



25. On the map, zoom in to Houston (or where you collected your points).
26. On the dashboard toolbar, click **Save** and choose **Save**.



27. Optionally, create more sample damage assessments using QuickCapture, Survey123, or Field Maps. Refresh your dashboard to see how the added samples change your dashboard.

Note: Alternatively, view the [sample dashboard](#) with hundreds of damage assessments.